

Bheema Sai Shankar Reddy Sabilla

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SUMMARY

Data Engineer with 3+ years of experience designing, building, and optimizing scalable batch and streaming data systems on cloud lakehouse platforms. Strong expertise in Python, SQL, PySpark, Azure Databricks, Delta Lake, and Microsoft Fabric, with hands-on experience delivering real-time analytics pipelines, governed data models, and enterprise-grade retrieval-augmented generation (RAG) systems.

PROFESSIONAL HIGHLIGHTS

- Designed and operated distributed data pipelines processing hundreds of millions of records with strict SLAs.
- Built production-grade lakehouse architectures supporting analytics, reporting, and machine learning workloads.
- Implemented event-time processing, backfills, and schema enforcement to ensure data correctness and reliability.
- Delivered AI-enabled data platforms integrating large language models with enterprise analytics foundations.

TECHNICAL SKILLS

- **Programming & Query:** Python, SQL, PySpark, Java, C, C++
- **Python Libraries:** Pandas, NumPy
- **Big Data:** Apache Spark, Spark Structured Streaming
- **Cloud:** Azure (Azure Data Factory, ADLS Gen2, Databricks, Event Hubs), AWS (S3, Glue), Microsoft Fabric
- **Lakehouse:** Delta Lake, Medallion Architecture, Snowflake
- **Analytics Engineering:** dbt (Core)
- **Data Modeling:** Star Schema, Slowly Changing Dimensions (SCD Type 1 & 2)
- **SQL:** Window Functions, Common Table Expressions (CTEs), Query Optimization
- **Data Engineering:** Data Aggregation, Data Reconciliation, Reporting Pipelines
- **BI:** Power BI, KPI Design, Semantic Models
- **Streaming:** Kafka, Event-Time Processing, Watermarking
- **Data Quality:** Schema Enforcement, Data Validation Frameworks, Data Observability, SLA Monitoring
- **Orchestration:** Apache Airflow, GitHub Actions, Git, CI/CD, Docker (Foundational)
- **LLM / RAG:** Retrieval-Augmented Generation (RAG), BM25, FAISS, Hybrid Retrieval, Cross-Encoders
- **Formats:** Parquet, JSON, CSV
- **ML Enablement:** Feature Engineering Pipelines, MLflow

EXPERIENCE

Data Engineer – AI/ML

Discover Financial, USA (Remote)

Expected May 2025 – Present

- Designed PySpark and Azure Databricks pipelines supporting fraud detection, credit risk, and marketing analytics.
- Built real-time ingestion pipelines using Kafka/Event Hubs and Spark Structured Streaming.
- Developed ML-ready feature datasets using Delta Lake with schema enforcement, lineage, and data quality checks.
- Implemented aggregation and reconciliation logic to ensure accuracy of downstream analytical reports.
- Partnered with data scientists to operationalize feature pipelines and align data outputs with business metrics.
- Optimized Spark workloads to improve performance and reduce compute costs through partitioning strategies.

Data Engineer

University of South Dakota, USA

Feb 2024 – Apr 2024

- Engineered Delta Lake pipelines processing 400M+ weekly records using PySpark.

- Implemented Medallion Architecture (Bronze–Silver–Gold) with automated validation and schema enforcement.
- Improved pipeline SLA from 93% to 99% through monitoring and reliability enhancements.
- Supported analytics and research reporting use cases with curated datasets.
- Translated stakeholder requirements into analytics-ready data models.

Data Engineer

Jun 2021 – Nov 2023

Hexaware Technologies, India

- Built and maintained enterprise ETL/ELT pipelines using Azure Data Factory, ADLS, Databricks, and Snowflake.
- Implemented CDC and incremental ingestion patterns supporting near-real-time analytics.
- Designed dimensional data models enabling self-service analytics and Power BI dashboards.
- Supported production pipelines through backfills, root-cause analysis, and performance optimization.
- Coordinated with cross-functional teams to meet delivery timelines and SLA commitments.

PROJECTS

S.H.A.R.P. – Enterprise Retrieval-Augmented Generation (RAG) Platform

- Built a hybrid RAG system combining BM25 and FAISS with hybrid score fusion.
- Implemented cross-encoder re-ranking and citation-grounded LLM generation using lightweight production-ready models.
- Achieved 64% Precision@5, 100% Recall@10, and approximately 90% citation accuracy.

Azure Databricks Lakehouse Optimization

- Optimized Bronze–Silver–Gold lakehouse layouts to improve analytics performance and cost efficiency.
- Applied Delta Lake optimization techniques including partitioning, compaction, and Z-ORDER.
- Improved data reliability through schema enforcement and pipeline monitoring.

Large-Scale Event Streaming Analytics

- Built real-time analytics pipelines using Spark Structured Streaming for high-volume event data.
- Implemented watermarking and late-arrival handling for accurate event-time aggregations.
- Persisted curated streaming aggregates in Delta Lake for downstream analytics.

EDUCATION

Master of Science in Computer Science

Jan 2023 – Dec 2025

University of South Dakota, USA

Bachelor of Technology in Computer Science

Jun 2017 – May 2021

Jawaharlal Nehru Technological University, India

CERTIFICATIONS

Google Data Analytics (Coursera) | Python for Data Science (IBM) | SQL for Data Science (Udemy) | Azure Databricks